

Abstract

Speech enhancement using time-frequency domain analysis often struggles with phase preservation, which is crucial for high-quality speech restoration. This paper proposes a novel Deep Learning model utilizing time-frequency coefficients. The method incorporates the simultaneous training of real and imaginary components without complex preprocessing, ensuring comprehensive enhancement of both magnitude and phase. The proposed CAR-UNet model has additional feature extraction layers which include ConvNeXT block and gated attention within the encoder, decoder, and skip connections, providing enhanced robustness and accuracy. The effectiveness of the approach has been rigorously validated using three clean speech databases LibriCorpus, TIMIT, and SUBAK.KO and one noise database DEMAND, involving ten types of noises in two different languages, namely Bengali and English. Under extreme noise conditions, such as -15dB SNR, our results demonstrate remarkable resilience. The resulting performance metrics significantly outperform state-of-the-art methods, establishing a new standard in speech enhancement.

Index Terms

CAR-UNet, phase enhancement, phase preservation, single channel speech, speech enhancement, STFT, time-frequency analysis, UNet

Full paper will be published soon!